SOLAX POWER

TOU Mode Operation Introduction

ENGLISH

V1.1 2025/1/17

Background

TOU is short for ‘TIME OF USE’. TOU function is added to better help users optimize the assets of home power stations for different scenarios. This function helps users set the operation strategy of site on weekends, weekdays. According to the regulations of the local grid, so as to achieve asset optimization better under the premise of complying with the regulations of the power company.

TOU Work Modes Introduction

Self-Use(by default):
PV energy is as self-sufficient as possible.
The priority of PV energy is: load>battery>grid connection. Minimum SOC: The minimum SOC in this mode is functionally the same as the on-grid minimum SOC. The two values do not interfere with each other. In the case of TOU, only this value is responded.
Min SOC: Functions similarly to Grid-Connected Min SOC, values do not interfere. In TOU scenarios, only this value is considered.

Battery hold:
The battery does not charge or discharge. PV energy is only for self-use or output. The battery is charged only when the SOC is lower than the set minimum SOC. When the minimum SOC is reached, the charging stops.

Peak-Shaving:
After the load power reaches the threshold, the battery will begin to discharge.
Threshold: Set the peak shaving limit, the battery is allowed to discharge after the imported power from the grid exceeds this value
Peak Limit: Sets peak shaving limit, allowing battery discharge when grid consumption exceeds this

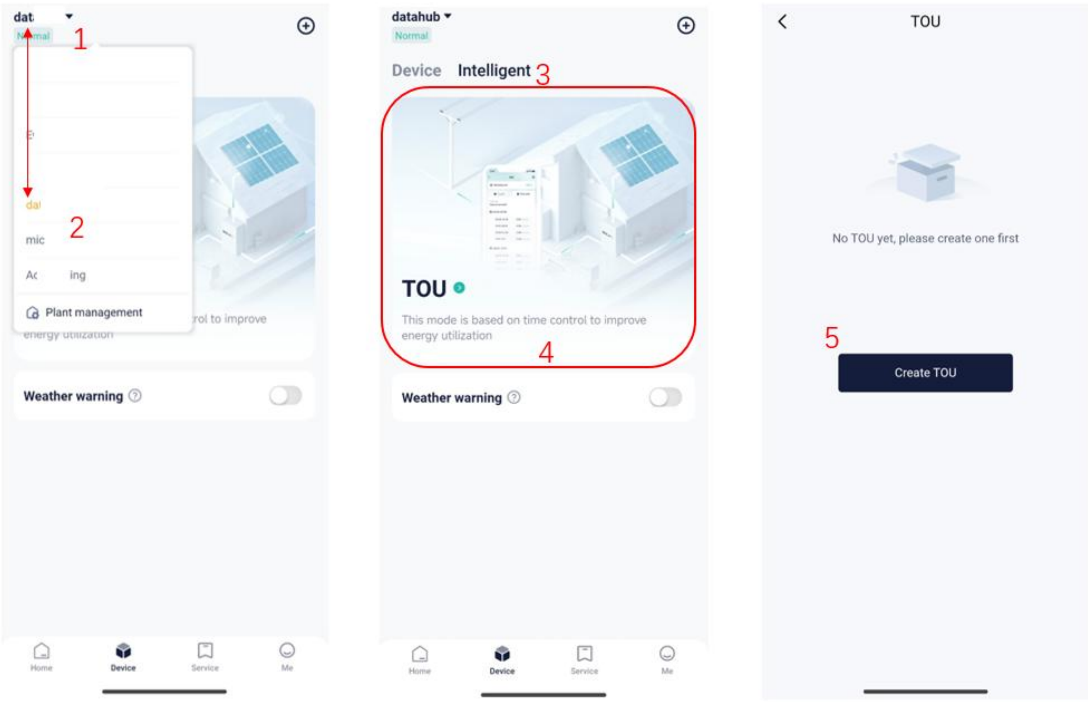
Charging:
PV will charge the battery as much as possible, and the user can choose whether to import from the grid
Allow Grid Power: Yes/No
Charge To: Target SOC for grid charging
Max Charge Power: Max power for charging with both PV and grid

Discharging:
The system will discharge at the "percentage" rated power until the "discharge threshold", and then switch to Self-use
AC Power: Max AC output power, can be % of rated or specified limit
Discharge To: Discharge Cutoff SOC

APP Settings (based on Cloud V6.3)

1. TOU main settings

Operation Instructions:

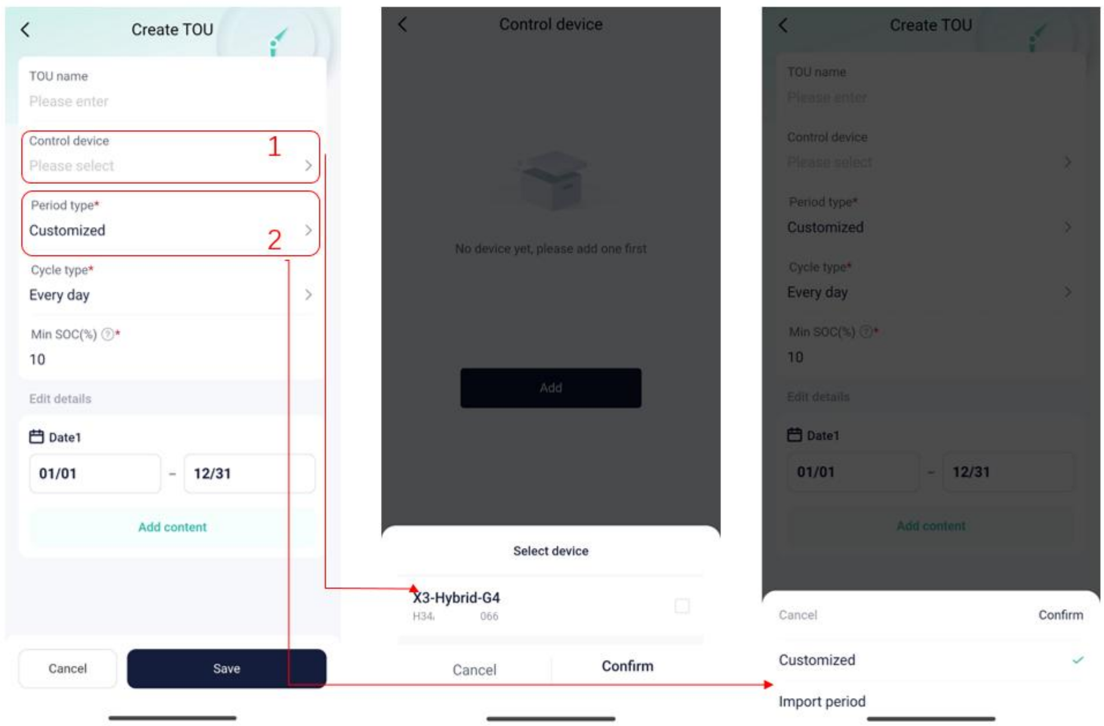


1.1 Select the Site 1.2 Entrance 1.3 Create TOU

①②Option to enable or disable TOU mode for this site

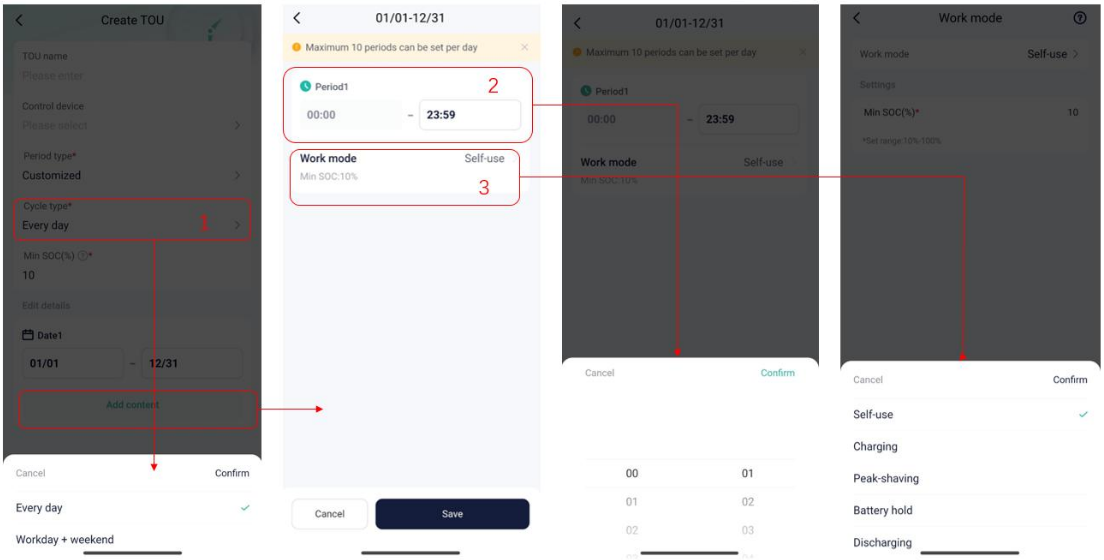
③④Click Intelligent, then Click TOU

⑤Create TOU



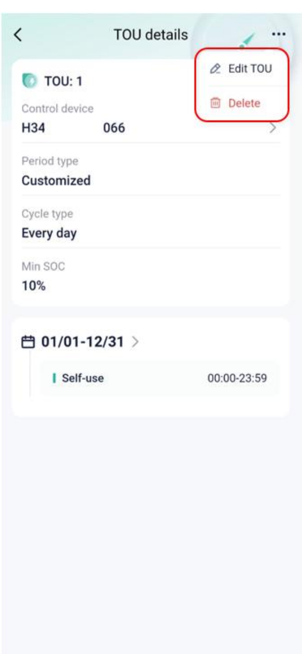
1.4 Fill in information 1.5 Select 1.6 Choose Type

①Select the device you want to edit
②Choose the Period Type

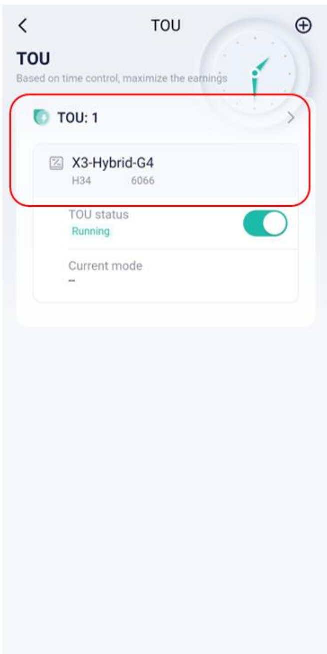


1.7 Select Cycle Type 1.8 Select Time 1.9 Choose Work Mode

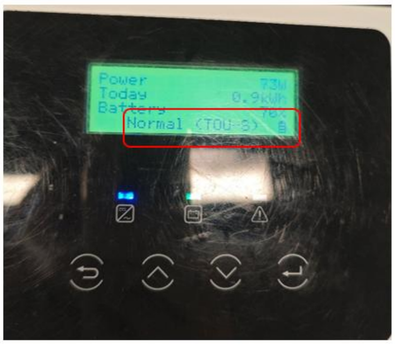
①Select Cycle Type to determine which TOU to use at different times
②③Add TOU and set different work modes for different time periods, and Min SOC be set at 10% by default



1.4 Edit TOU



1.5 Set TOU successfully

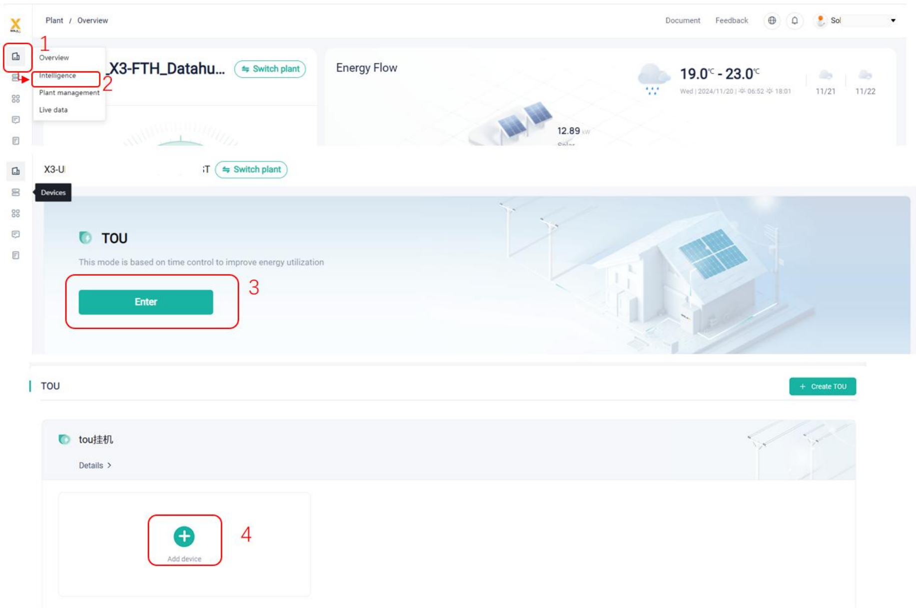


1.6 Displayed on the inverter screen

Web Settings (based on Cloud V6.3)

1. TOU main settings

1.1 Entrance



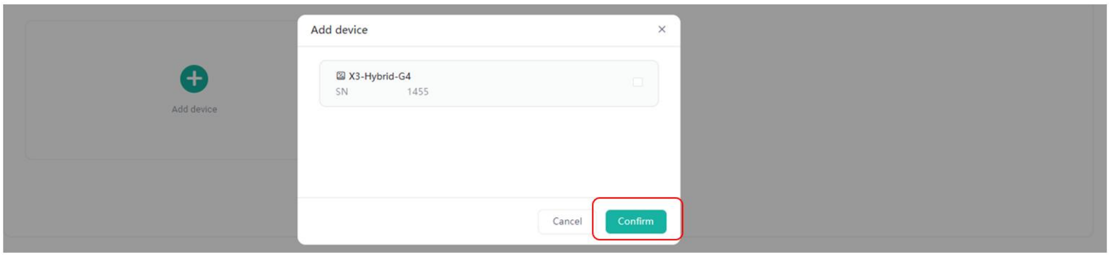
Operation Instructions:

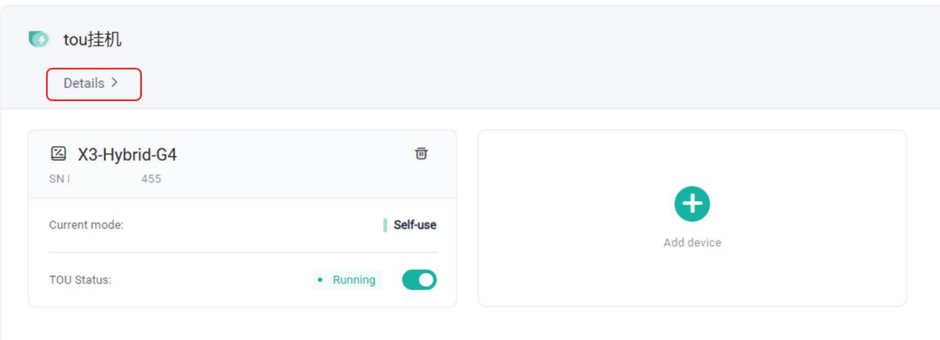
① In the navigation bar on the left, select the 'device'

② Click the 'Intelligence' in the red circle

③④ On the TOU panel, click 'Enter'. On the page that is displayed, click 'Add Device'.

1.1.2 Add device

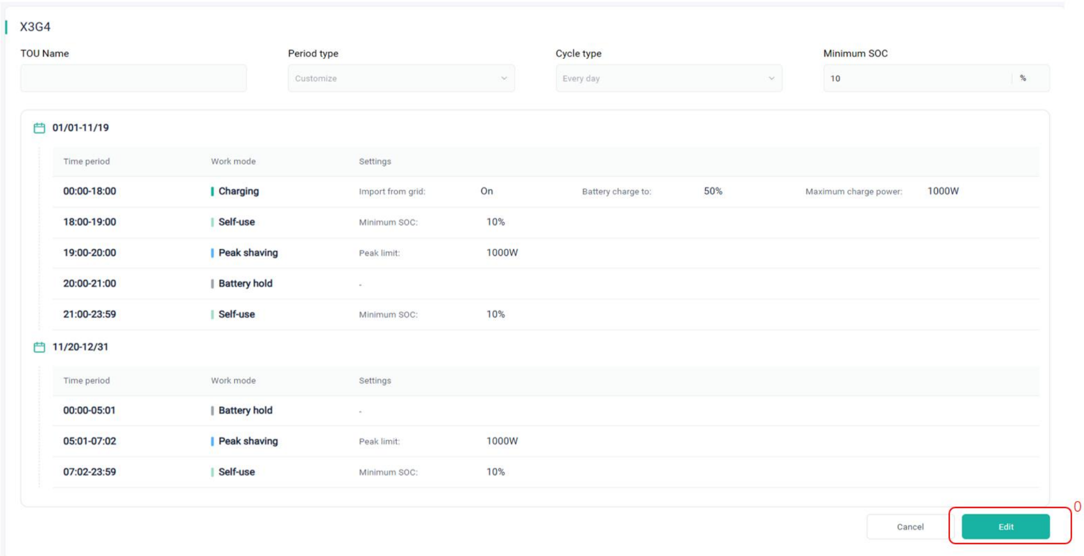


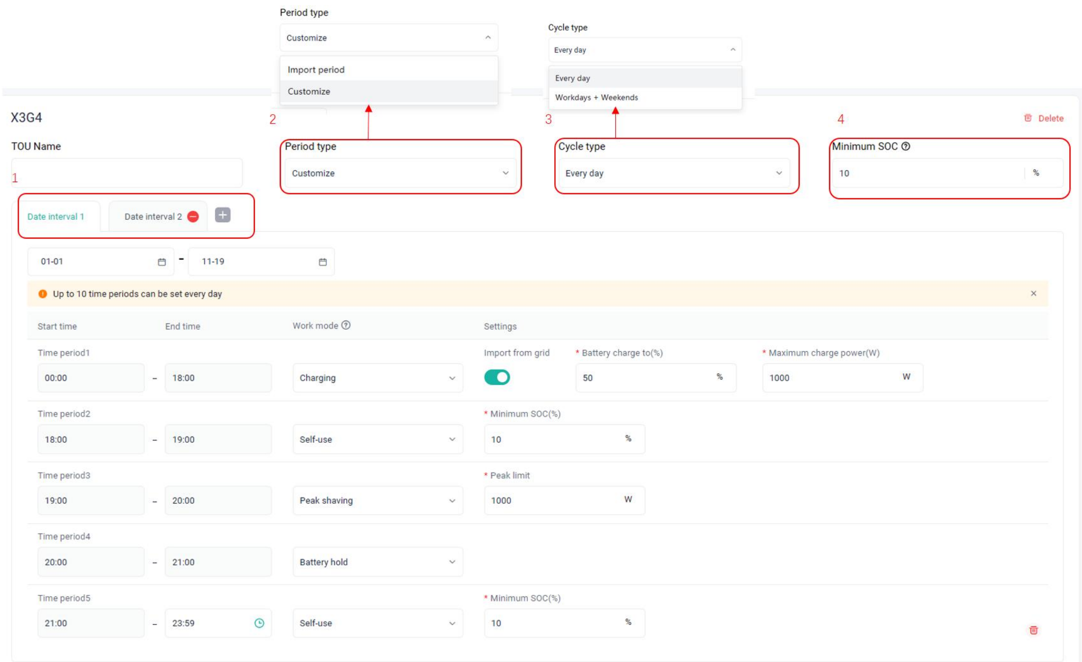


Operation Instructions:

Click the 'Confirm', then click the 'Details' to check more information.

1.2 Fill in the information and rules





Operation Instructions:

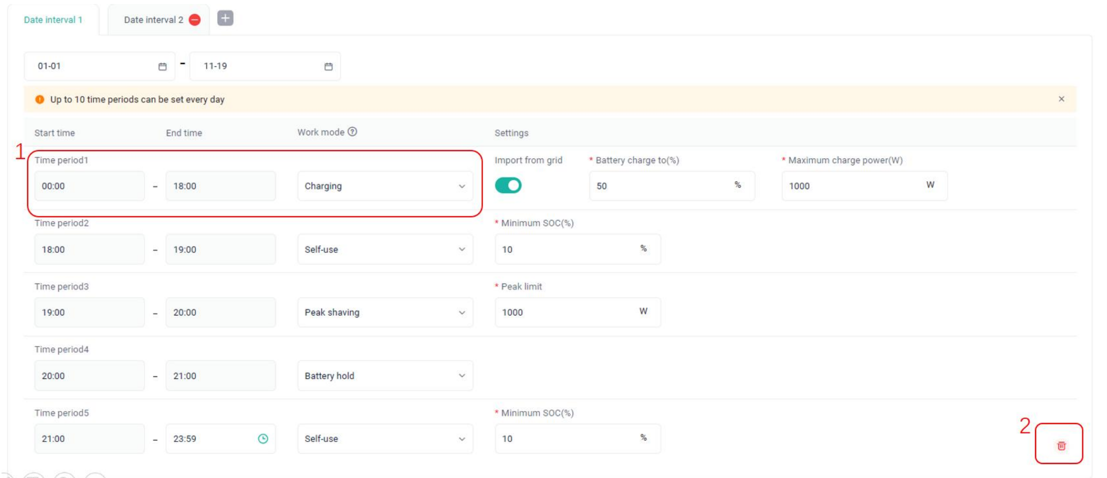
⓪ Click the ‘Edit’ if you want to alter your settings.

① Different date intervals can be set here, and a maximum of ten can be set

② You can choose to import the entire set directly, or it can be set by yourself

③ The cycle type can be selected as either "Everyday" or "Workdays + Weekends".

④ You can set the ‘Minimum SOC’ with a default setting of 10%.

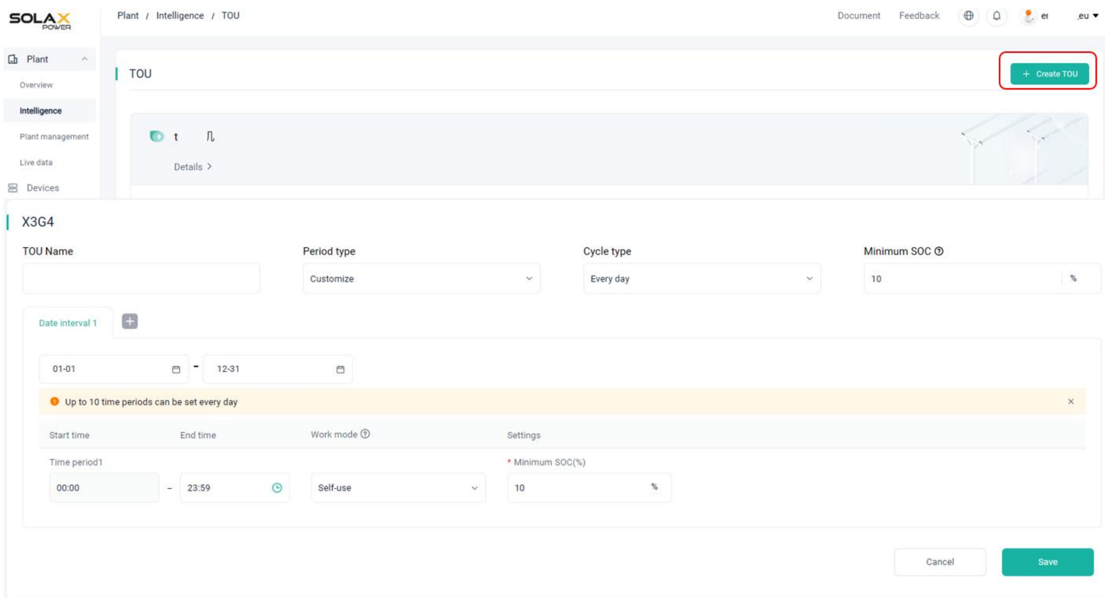


Operation Instructions:

① You can set the work mode for specific time periods here.

② Here you can delete.

1.3 Add TOU rules



Operation Instructions:

① Click Create New TOU and fill in the same content as before.

Compatibility

Inverter:

Function	Inverter Type
TOU Function	X1-IES
	X3-IES / X3-IES-P
	X1-Hybrid-G4
	X3-Hybrid-G4
	X1-Fit-G4
	X3-Fit-G4
	X1-Vast 5-10kW
	X3-HYB-G4 PRO
	X3-ULTRA
	A1-Hybrid-G2
	A1-AC-G2
	J3-ULTRA-LV
	J3-ULTRA

Besides the inverter, the dongle's hardware and firmware also need to be compatible, as listed below:

Function	Pocket Type	Firmware Version
Unlimited push	Pocket WiFi 2.0/Plus(SX/SJ)	≥ 3.012.01
	Pocket WiFi 3.0(SV/SR)	≥ 3.012.01
	Pocket WiFi 3.0 (SS)	> 3.037.04
	Pocket LAN 2.0(SL)	≥ 3.006.04
	Pocket LAN 3.0(SY)	≥ 3.006.04
	Pocket WiFi+LAN(SN)	≥ 1.002.09
	Pocket WiFi+4GM(SM)	≥ 1.002.14